

A Heat Exchanger Simulation

Simulating a shell-and-tube heat exchanger for industrial applications

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Abstract — Heat exchangers have been playing an important role in many solar power plants, by transferring energy between working oil to molten salt, enabling efficient storage of solar energy. Simulating shell-tube heat exchangers is therefore crucial in understanding the heat transfer and fluid dynamics phenomena, allowing better understanding and comparison between different designs and setups. A shell-and-tube geometry and a physical model describing fluids and the regarding dynamics and coupled heat transfers is presented as well as simulations made with *SimFlow* and the underlying solution *chtMultiRegionFoam* from *OpenFOAM*, aiming to understand current industrial applications and research.

1 Introduction

For a realistic simulation of the heat transfer between two fluids in a shell-tube heat exchanger, the coupling of the governing fluid dynamic equations and heat transfer must follow from some essential assumptions made on the system, as particular boundary conditions and the specific fluid and solid properties. Therefore, the underlying physical effects defined by governing equations can be specified for the particular use case and then solved upon the choose of an appropriate solver. The shell-and-tube heat exchanger was considered for this simulation once it represents the most common type of heat exchanger in practice (Abeykoon, 2019), where 2 different fluids flow physically separated on two different sections of the heat exchanger. The goal is then to produce a precise mathematical model that can properly be simulated using, for example, the *chtMultiRegionFoam* solution from *OpenFOAM*, respectively, and further tools as *SimFlow*, considering the underlying fluid dynamics and solid heat conduction, with conjugate heat transfer between solid and fluid regions buoyancy effects. The presented research can be split in two main sections: an experimental scenario, where water flows in both the shell and the tube of the heat exchanger and an industrial application, focusing on solar power plants usage.



Figure 1 Heat exchangers in *Planta Solar de Almería*

For this experimental setup, baffled and non-baffled geometries were simulated, as well as different inlet velocities on the shell side. For the second scenario (and seeking a simulation representing a real word application), molten-salt was considered to flow on the shell side, while thermal oil on the tubes. This setup differs from the first one by showing thermophysical properties desired on shell-and-tube heat exchangers, as the ones existing in the *Planta Solar de Almería*, as in figure 1 (Rodríguez-García et al. (2013)). With this setup, the model aims to be applied on a simulation representing the physical effects on the heat exchange from the warm fluid to the cold. Starting with the essential assumptions regarding the materials and fluids, followed by the following setup for the two different scenario: the the governing equations for physical events that take place. Boundary conditions, initial conditions and thermophysical properties are then prescribed, in order to provide a complete setup for the respective simulation. After prescribing the physical model, a description of the numerical schemes and solvers used is presented. Finally, the results for the different setups are shown and discussed.

2 Assumptions

With regards to the general fluid properties, both fluid flows on both setups share the following assumptions:

- A1 The flow is continuous.
- A2 The flow is Newtonian.
- A3 The flow is isotropic.
- A4 The fluids are incompressible.

3 Governing equations

3.1 Equilibrium equations

In order to be able to understand the fluid dynamics given the assumptions above, the following set of equations must be solved for each of the fluid regions. The continuity equation and Cauchy's first law of motion are, respectively (Greenshields and Weller, 2022):

$$\frac{D\rho}{Dt} + \nabla \cdot (\rho u) = 0$$

$$\rho \frac{\partial u}{\partial t} + \nabla \cdot (\rho u u) = \nabla \cdot \sigma + \rho b$$

Here ρ represents the fluid density, u is the velocity, b represents any body force per unit mass and σ is the Cauchy stress tensor.

3.1.1 Non-turbulent transient setup

The importance of a transient simulation arises from the urge to understand how changes in an equilibrium state (e.g., changes on the inlet temperature or fluids inlet velocity) impacts the system. Moreover, it also allows to understand how different regions of the heat exchanger are affected differently by these changes. Turbulence adds computational complexity to the numerical methods, as well as the time dependent system model (Rumsey et al., 1996). From the given assumptions, as well as discarding the effects of gravity and external forces, the continuity equation and the momentum equation can be simplified according to Laitinen et al. (2020) as:

$$\nabla \cdot u = 0 \quad (1)$$

$$\rho \left(\frac{\partial u}{\partial t} + \nabla \cdot (uu) \right) = -\nabla p + \mu \nabla^2 u \quad (2)$$

The energy equilibrium due to the heat exchange at the interface between the fluids at the shell or tube and the solid can be solved according to Laitinen et al. (2020):

$$\frac{\partial T}{\partial t} + \nabla \cdot (uT) - \frac{\kappa_f}{\rho_f c_{p,f}} \nabla^2 T = 0 \quad (3)$$

which can be simplified in the solid side to:

$$\nabla^2 T = \frac{\rho_s c_{p,s}}{\kappa_s} \frac{\partial T}{\partial t}$$

where T represents the temperature.

3.2 Heat transfer

Besides the isolated systems for the 2 fluids, the conservation of energy at the interface must hold, which means:

$$\kappa_f \frac{\partial T_f}{\partial n} = \kappa_s \frac{\partial T_s}{\partial n}$$

and

$$T_f = T_s$$

Under this context, κ refers to the respective thermal conductivity, c_p to the specific heat capacity, $\frac{\partial}{\partial n}$ to the derivative normal to the interface and the indexes s and f are respectively solid and fluid.

4 Methodology

4.1 Geometry and setup

In order to proceed to a correct simulation it is still necessary to define the suitable boundary conditions, initial conditions and thermophysical properties for the geometry presented in figure 2, which was provided by the fluid dynamic simulation tool *SimFlow* and different fluid configurations for the tube side (identified by A) and the shell (identified by B). The interface between these fluids was considered to be a infinitely small solid layer. Moreover, the impact of baffles (Figure 3) inside the shell were considered for the study.

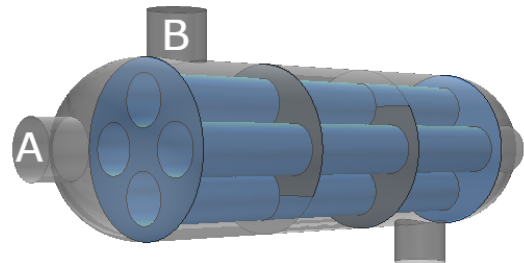


Figure 2 Geometry Profile

Table 1 presents a list of general boundary conditions that are applied to all the setups (i.e. are general boundary conditions), following the assumptions that external walls, baffles and tube and shell outlets are

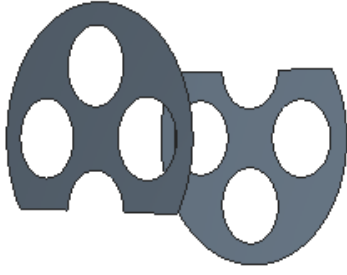


Figure 3 Baffles

adiabatic and the flux follows a *homogeneous Neumann* boundary conditions. Moreover, it is also considered that at the contact walls between the tubes and the shell *No-Slip* and *Coupled Temperature* boundary conditions apply and that the gravitational effects are not taken into account for the simulation.

Boundary	Velocity	Temperature
Tube outlet	$\frac{\partial u}{\partial n} = 0$	$\frac{\partial T}{\partial n} = 0$
Tube inner wall	$u_{patch} = 0$	Coupled
Tube outer wall	$u_{patch} = 0$	Coupled
Shell outlet	$\frac{\partial u}{\partial n} = 0$	$\frac{\partial T}{\partial n} = 0$
Shell wall	$u_{patch} = 0$	$\frac{\partial T}{\partial n} = 0$

Table 1 General Boundary Conditions

4.2 Water - Water

Firstly, three configurations where water flows on both the shell and the tube were considered. Here the difference between these configurations is the fluid inlet velocity (u_{in}) at the shell side. Hence, the initial conditions are the same for the three different configurations and are presented on table 3. It is also necessary to state the material and physical properties of both the water at shell-side and the water on the tube, presented in table 2. It is further assumed that the following thermophysical properties are independent of the temperature. The studied temperatures and thermophysical properties follow from Abeykoon, 2019.

Lastly, the specific boundary conditions are presented in table 4.

For the three different configurations is it considered, respectively, $u_{in} = 0.01$, $u_{in} = 0.03$ and

	Physical Property	Unit	Quantity
Water	Density	kg/m^3	995.7
Shell	Specific Heat	$J/kg K$	4197
	Viscosity	$Pa \cdot s$	7.98×10^{-4}
	Thermal Conduc.	W/mK	0.615
	Prandtl Number	-	5.43
Water	Density	kg/m^3	971.8
Tube	Specific Heat	$J/kg K$	4197
	Viscosity	$Pa s$	3.55×10^{-4}
	Thermal Conduc.	$W/m K$	0.670×10^{-3}
	Prandtl Number	-	2.22

Table 2 Thermophysical Properties

Cell Zones	Velocity	Temperature
Shell fluid	$u = 0$ m/s	$T_0 = 302.8K$
Tube fluid	$u = 0$ m/s	$T_0 = 302.8K$

Table 3 Initial Conditions | Water - Water

$u_{in} = 0.05$ [m/s]. These velocities follow from the interest on simulating the heat exchanger from three different Reynolds number at the inlet shell, respectively $Re = (1000, 3000, 5000)$. These values represent a laminar, transitional and turbulent flow scenario at the inlet shell and follow from the following relationship (Spurk and Aksel, 2020):

$$Re = \frac{\mu}{\rho d} u \quad (4)$$

where $d = 0.08$ [m] is the diameter from the inlet at the shell. Moreover, to a better understanding of the result, we also introduce the heat transfer rate, as defined in Wang et al., 2009 :

$$Q = W_h c_{p,f} \Delta T \quad (5)$$

where W_h represents the mass velocity, that can be computed as $W_h = \rho u_{outlet}$ and $\Delta T = T_{outlet} - T_{inlet}$.

Boundary	Velocity	Temperature
Tube inlet	$u = 0.1 \text{ m/s}$	$T = 353.15 \text{ K}$
Shell inlet	$u = u_{in} \text{ m/s}$	$T = 302.8 \text{ K}$

Table 4 Specific Boundary Conditions | Water - Water

4.3 Molten salt - Oil

The simulation of a molten-salt heat-exchanger is important, as these type of heat-exchanger play a significant role in multiple industrial applications (e.g. solar plants (Rodríguez-García et al., 2014)). The chosen heat transfer fluids used on the heat exchanger heavily impacts its efficiency, hence representing and extensive research area (Benoit et al., 2016)(Vignarooban et al., 2015). One of the most common molten salt on solar plant facilities is the so called *solar salt*, composed of 60% NaNO_3 and 40% KNO_3 and presenting an operating temperature range of $[535.15 - 600]$ [K]. As for the thermal oil a common choice is the so called *Therminol VP-1* (*Therminol VP-1 Ultrahigh-temperature, Vapor/Liquid Phase Fluid 12° to 400° C*, n.d.).

On this simulation the solar salt runs on the shell side, identified by letter B in figure 2, while hot oil flows on the tubes, identified by letter A. Here, five different setups based on the shell inlet velocity are considered. Specifically, the scenarios regarding $u_{in} = \{0.005, 0.01, 0.03, 0.05, 0.079\} \text{ [m/s]}$. A further scenario where the thermal resistance of the layer between the two fluids is considered. For this, it is considered that the layer is made out of copper, with a thickness is $\delta = 5 * 10^{-03} \text{ [m]}$ and the thermal conductivity $\kappa = 385 \text{ [W/mK]}$ (Young, 1992). The thermo-physical properties for both solar salt and thermal oil are shown in table 5, according to Benoit et al., 2016

Cell Zones	Velocity	Temperature
Shell fluid	$u = 0 \text{ [m/s]}$	$T_0 = 563.15 \text{ [K]}$
Tube fluid	$u = 0 \text{ [m/s]}$	$T_0 = 563.15 \text{ [K]}$

Table 6 Initial Conditions | Molten Salt - Oil

Lastly, the specific boundary conditions are presented in table (7)

Boundary	Velocity	Temperature
Tube inlet	$u = 0.1 \text{ [m/s]}$	$T = 646.15 \text{ [K]}$
Shell inlet	$u = u_{in} \text{ [m/s]}$	$T = 563.15 \text{ [K]}$

Table 7 Specific Boundary Conditions | Molten Salt - Oil

Here, the considered properties regarding the molten salt and oil

4.4 Numerical Methods

The simulations presented in this study were based on the *OpenFOAM* solution available through *SimFlow* Technologies, n.d. package for conjugate heat transfer: *chtMultiRegion*.

4.4.1 chtMultiRegionFoam

The *chtMultiRegionFoam* is a solution from *OpenFOAM* (Weller et al., 1998), an open-source software based on the finite-volume method for partial differential equations. This solution provides a numerical approach to the problems considering conjugate heat transfer between regions, with the possibility to consider solid heat conduction, buoyancy effects, turbulence, reaction and radiation modelling (Greenshields and Weller, 2022). For the problem considered on this research, there are no solid regions defined as cell regions (as the interface layer from the heat exchanger is considered to be infinitesimally thin) and therefore, the only two different cell regions where the solution is applied are the 2 fluids. Hence, the overview of the solution as applied to the considered problems can be understood as shown in figure 4, with a *PIMPLE* loop taking place sequentially for both fluids, for each time step.

The *PIMPLE Foam* solves for each time step the energy equation, taking a prediction for the momentum through the momentum equation and iterative correcting it with the pressure equation, as it is expressed in figure 5. With regards to discretization for time, convection, gradients and interpolations terms, the schemes used can be observed under table 8. To solve the linear systems, the linear solvers presented in table

	Physical Property	Unit	Quantity
Molten Salt	Density	kg/m^3	2263.7
	Specific Heat	$J/kg \cdot K$	$1396.018 + 0.172T$
	Viscosity	$Pa \cdot s$	$0.085681 - 2.8092 \times 10^{-4}T + 3.4882 \times 10^{-7}T^2 - 1.4740 \times 10^{-10}T^3$
	Thermal Conduc.	$W/m \cdot K$	$0.391 + 1.9 \times 10^{-4}T$
	Molecular Weight	g/mol	85
Thermal Oil	Density	kg/m^3	1191
	Specific Heat	$J/kg \cdot K$	$716 + 2.82T$
	Viscosity	$Pa \cdot s$	$2.02 * 10^{-4}$
	Thermal Conduc.	$W/m \cdot K$	$0.133743 - 0.81948 \times 10^{-5}T - 1.9226 \times 10^{-7}T^2 + 2.5034 \times 10^{-11}T^3 - 7.2974 \times 10^{-15}T^4$
	Molecular Weight	g/mol	166

Table 5 Thermophysical Properties

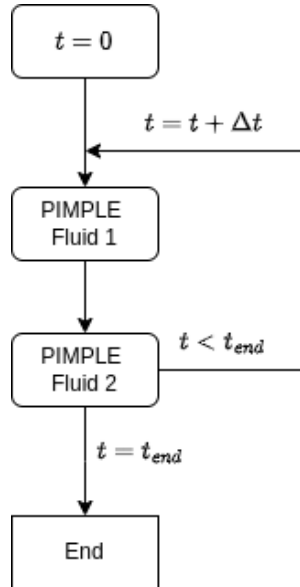


Figure 4 *chtMultiRegionFoam* Overview

Discretization	Scheme
Time	Implicit-Euler
Convection	1st Order Upwind
Gradients	Gauss Linear
Interpolation	Linear

Table 8 Discretization schemes

Once we assume fluid incompressibility (A4), equations do not need to be solved with respect to density. Hence the solver only needs to be specified for the application of the *SimFlow* software package, but is not used under the numerical schemes. The choice of the GAMG (Geometric agglomerated Algebraic MultiGrid), PCiCG (Preconditioned bi-conjugate gradient) Stabilized with DILU and PCG (Preconditioned Conjugate Gradient), follow from the suggestion made by *SimFlow* for the respective vector field. These suggestions can be confirmed through Greenshields and Weller, 2022, as GAMG suits applications to vector fields with large range of influence (e.g. as pressure) and the conjugate gradient methods follow a "descent" trajectory to the solution of a minimization problem, considering the symmetry/asymmetry

Variable	Solver	Tolerance
Pressure	GAMG	10^{-6}
Velocity	PBiCG Stabilized	10^{-6}
Enthalpy	PBiCG Stabilized	10^{-6}
Density	PCG	10^{-6}

Table 9 Linear Solvers

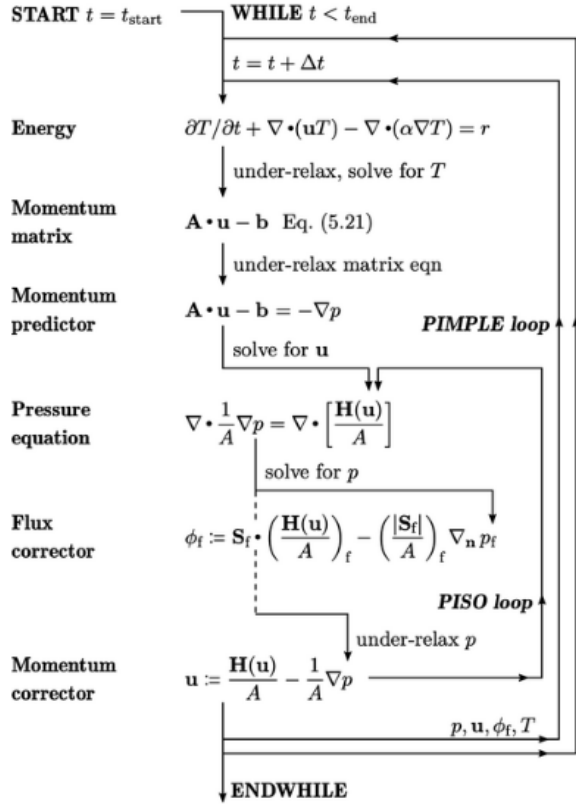


Figure 5 PIMPLE Foam

of the system matrix distinguish by the application conjugate/bi-conjugate scheme, respectively.

5 Discussion

5.1 Transient

5.1.1 Water - Water

Simulation results for the initial and boundary conditions specified in tables 3 and 4 are presented on figures 6, 7 and 8. Figure 6 depicts the temperature evolution at the outlet (averaged through the outlet area) over 900 seconds. For each pair Baffles/No baffles, the following results can be highlighted: baffled systems show a higher temperature increase, for the same inlet shell velocity; lower inlet velocities lead to a lower temperature increase at the shell outlet. The maximum temperature increase was for the baffled system with $u_{in} = 0.01[m/s]$, with shell outlet temperature reaching 318K. However, if the fluid motion is taken into account by considered the heat transfer coefficient introduced in equation 5, we can see that even though the pattern that baffled systems present a higher heat increase at the outlet, taking in play the fluid velocity

at the outlet, the system with higher inlet shell velocity show a higher heat transfer coefficient, followed by the one with medium and low shell inlet velocity, respectively.

5.1.2 Molten Salt - Oil

The results of the molten salt and thermal oil simulations for the specified thermophysical properties, initial and boundary conditions on tables 5, 6, 7 are presented in figures 9, 10 and 11. Similarly to the results presented in the section 5.1.1, figure 9 depicts the temperature evolution at the outlet (averaged through the outlet area) over 900 seconds, while figure 10 the heat transfer rate. Here the same pattern is observed: higher shell inlet velocities lead to lower averaged outlet temperature whilst the heat transfer coefficient being greater for higher shell inlet velocities. With regards to the analysis between the scenario considering resistance of the layer between the two fluids, figure 11 depicts no noticeable change on the behaviour of the shell outlet averaged temperature by considering thermal resistance. Comparing these results with the ones obtained for Water-Water and discarding that the initial temperature difference between the fluids at the *Molten Salt - Oil* against *Water - Water* setup is higher at the molten salt configuration ($\Delta T_{MSOil} > \Delta T_{Water}$), the *Molten Salt - Oil* setup presents generally lower heat transfer rates. However, due to the capacity and chemical stability to deal with higher temperatures, this setup has been a choice with industrial applications as in the *Planta Solar de Almeria*.

6 Conclusion

The present study aimed to understand and simulate the complexity of modeling a shell-and-tube heat exchanger and the different features, numerical and physical, that impact the simulation. Understanding the physical problem was crucial to correctly simulate the physical phenomena, as well as dominating the existing software and solvers for the complex model studied under this study. With the solver *chtMultiRegionFoam* made available by the open-source software *OpenFOAM* it was possible to tackle the problem considering the different features required for this simulation, with fully control and awareness of the available solution. *SimFlow* as shown to be a powerful tool to simulate this model: it combines the powerful and well-studies numerical for finite volume methods from *OpenFOAM* with a user-friendly interface, making it

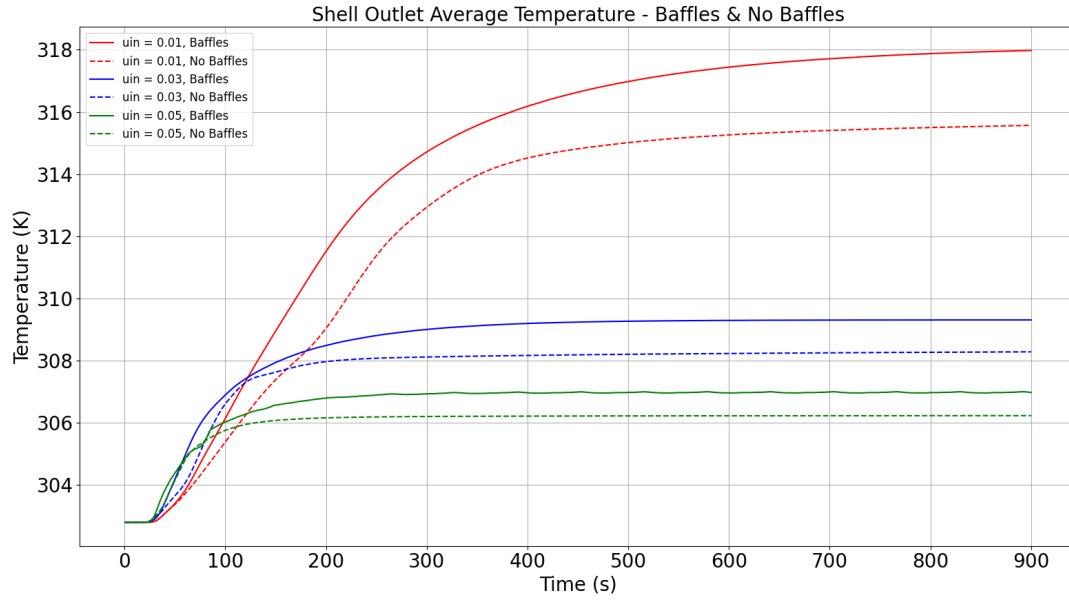


Figure 6 Average Outlet Temperature | Water - Water

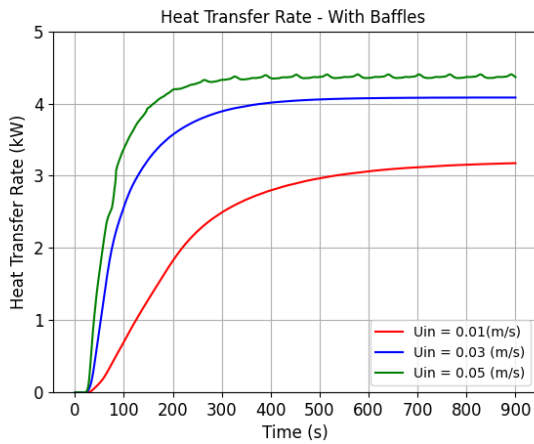


Figure 7 Heat Transfer Rate With Baffles | Water - Water

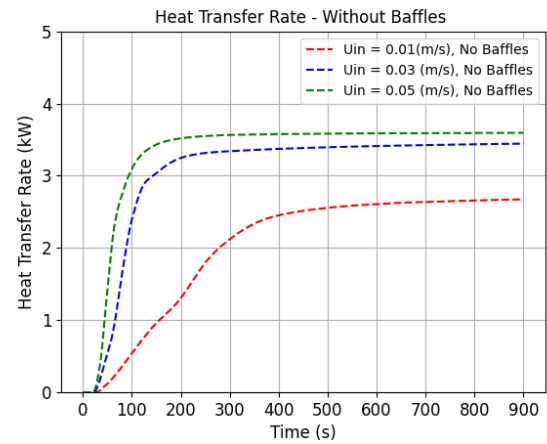


Figure 8 Heat Transfer Rate Without Baffles | Water - Water

easier to understand and apply the open-source tools. Nevertheless, the arising limitation due to commercial license availability can become a hindrance for more complex geometries and meshes. With the water-water simulation, it was shown that baffles improve the efficiency of the heat exchanger, leading to higher heat transfer coefficient, regardless of the shell inlet velocity considered. Furthermore, the impact of the chosen shell inlet velocity can produce different outcomes, depending on the type of application: if the goal of the heat exchanger is to increase the temperature absolute value, slower inlet shell flows show to be a better application, whereas if the interest is in producing a higher heat transfer (i.e., considering the mass flow),

higher velocities shown to be a better solution (producing higher heat transfer rates). Similar results have been obtained with the *Molten Salt - Oil*, however on a different magnitude scale.

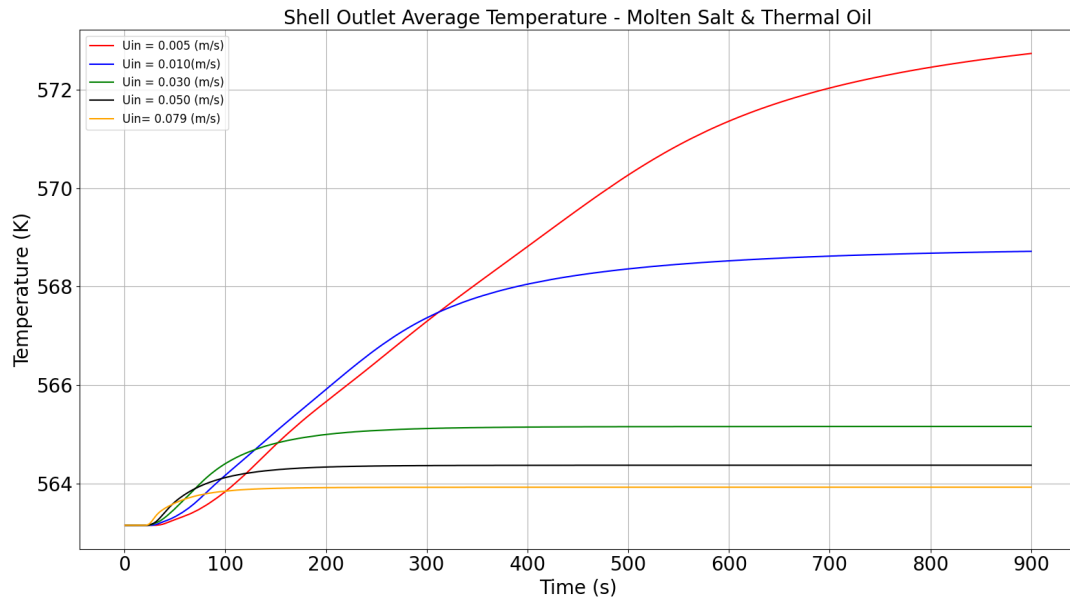


Figure 9 Average Outlet Temperature | Molten Salt - Oil

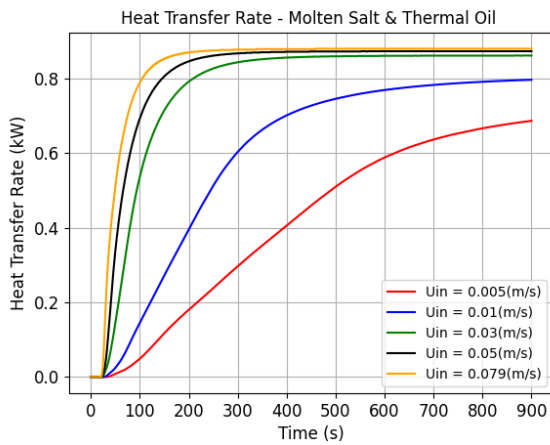


Figure 10 Heat Transfer Rate | Molten Salt - Oil

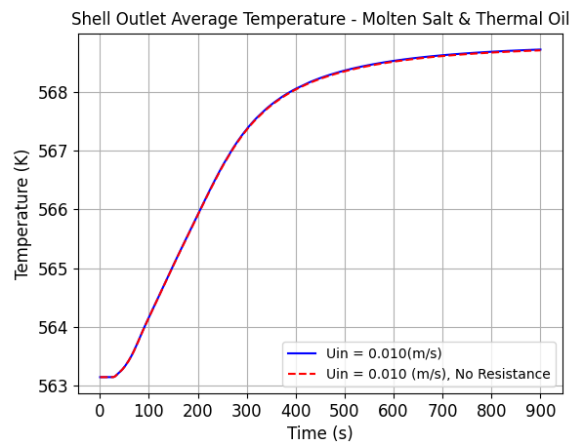


Figure 11 Average Outlet Temperature | Molten Salt - Oil | Resistance - No Resistance

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Appendices

A Turbulent steady-state

Steady-state analysis focus on understanding the system properties as the temperature distribution and swirl distributions when it reaches, for example, an equilibrium state. However, when considering turbulence model for the fluid dynamics, the complexity of the system analysis increases and further strategies are developed to approach the capture of the physical dynamics. For this research, Reynolds-Averaged Navier Stokes were considered. These methods are based on the *Reynolds Equations* (Spurk and Aksel, 2020). These consider an average form of the variables as *random quantities* in fully developed turbulent flows (after the transition phase has been completed). For the velocity field, this means

$$u_i(x_j, t) = \bar{u}_i(x_j, t) + u'_i(x_j, t)$$

As the considered setup is *statistically steady-state*, this means that the time-mean value can be calculated as (Spurk and Aksel, 2020):

$$\bar{u}_i(x_j) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{t-T/2}^{t+T/2} u_i(x_j, t) dt$$

Given this, the continuity, momentum and energy balance equations must also be updated Pal et al., 2016:

$$\begin{aligned} \frac{\partial \bar{u}_i}{\partial x_i} &= 0 \\ \rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} &= -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \tau_{ij} \\ \frac{\partial (T \bar{u}_j)}{\partial x_j} &= -\frac{\partial}{\partial x_j} \left(\left(\frac{\nu}{Pr} + \frac{\nu_t}{Pr_t} \right) \frac{\partial T}{\partial x_j} \right) \end{aligned}$$

Following an approach from Pal et al. (2016) the simulations were performed according to standard $k-\epsilon$ model, leading to 2 additional equations. Turbulence kinetic energy and its dissipation must additionally be solved for each system. Considering the equation for turbulent kinetic energy k comes as:

$$\begin{aligned} \rho \bar{u}_j \frac{\partial k}{\partial x_j} &= \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_T}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] \\ &\quad + \tau_{ij} \frac{\partial \bar{u}_i}{\partial x_j} - \rho \epsilon \end{aligned} \quad (6)$$

The dissipation rates ϵ with respect to k is respectively:

$$\begin{aligned} \rho \bar{u}_j \frac{\partial \epsilon}{\partial x_j} &= \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_T}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] \\ &\quad + C_{\epsilon 1} \frac{\epsilon}{k} \tau_{ij} \frac{\partial \bar{u}_i}{\partial x_j} - C_{\epsilon 2} \frac{\epsilon^2}{k} \end{aligned} \quad (7)$$

where

$$\tau_{ij} = \mu_T \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} - \frac{2}{3} \frac{\partial \bar{u}_k}{\partial x_k} \delta_{ij} \right) - \frac{2}{3} \rho_k \delta_{ij}$$

$$\mu_T = \rho C_\mu \frac{k^2}{\epsilon}$$

where τ_{ij} represents the Reynolds stress tensor, $\bar{u}_i$ is the average velocity component and μ_T the eddy viscosity. Here, the parameters also follow as in Laitinen et al. (2020) and Laitinen et al. (2020). Hence, $C_\mu = 0.09$, $C_{\epsilon 1} = 1.44$, $C_{\epsilon 2} = 1.92$, $\sigma_k = 1.0$ and $\sigma_\epsilon = 1.3$.